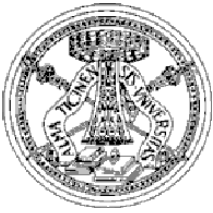


Health Care case study

Part 2 – Project assignment

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R 251017



Introduction

Project Scope

Requirements

Architecture

Assignment

Project Management

Integration

Time

Quality

Communi-
cations

Procurement

Scope

Cost

HR

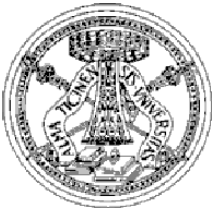
Risk

Stakeholder

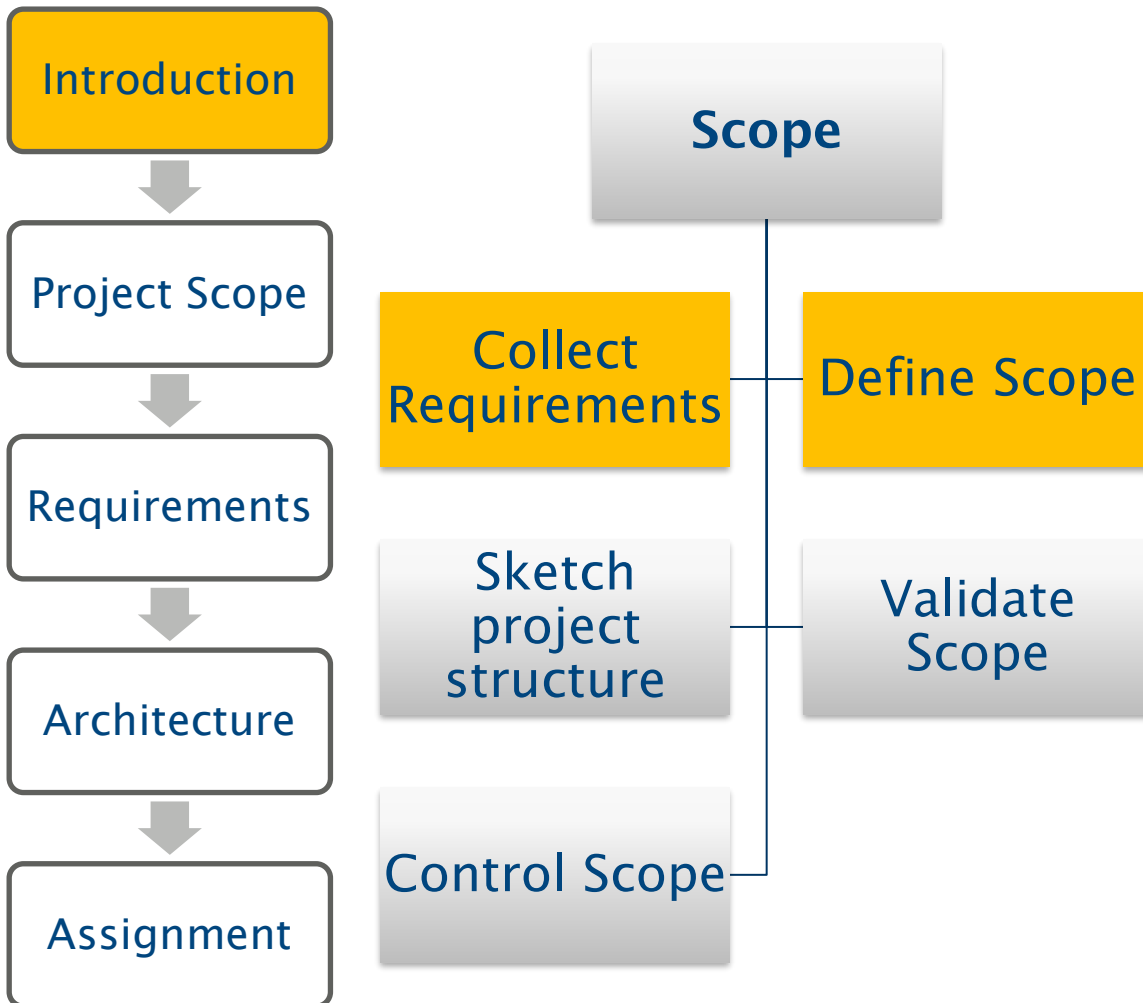
Project scoping in PMBOK

PMBOK's scoping includes:

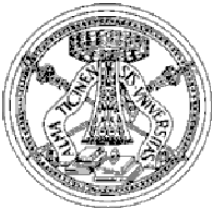
1. Collect Requirements
2. Define Scope
3. Create WBS (= the project structure)
4. Validate Scope
5. Control Scope



Project scoping assignment



- Our assignment covers two processes of Project Scope
 - Collect Requirements
 - Define Scope
- To better fit the profile of Software Projects
 - We first define the project scope
 - Then we model requirements within the project scope



Introduction

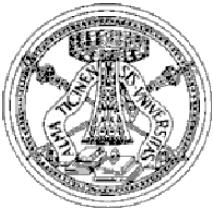
Project Scope

Requirements

Architecture

Assignment

- HC assignment requires student team to develop a scoping document for the HC case.
- The scope can address the whole system or a sub-system (a “theme” in Agile jargon).
- In any case, scoping should be consistent with
 - As-is situation, described in part 1 of the HC case study;
 - To-be scenario, summarized in the subsequent pages.



Project scoping assignment for HC

To-be scenario: user side

Introduction

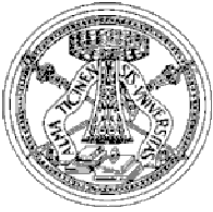
Project Scope

Requirements

Architecture

Assignment

- A patient can book easily any visit on the smart phone or on any counter or reservation center
- Both GP and patient can easily track the itinerary of the prescription in real time and change/cancel the reservations /appointments for visits /exams /operations.
- GPs should be supported by a knowledge base / machine learning app that advises them about similar cases and lists references.



Project scoping assignment for HC

To-be scenario: development implications

Introduction

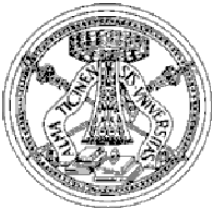
- An ideal scenario requires new apps to be developed, each reflecting the behavior of a user class
 - GP app, to manage the whole life cycle of prescription.
 - Patient's app, by which he/she can easily book visits, access prescriptions and reservations, and pay the ticket
 - For the patients, mock-up on the website to illustrate how that software will work
 - For the Region managers, a business intelligence app, to measure the service times and provide real-time statistics.
 - For patients and stakeholders, a reputation app, like Trip Advisor, where patients express their feedback.

Project Scope

Requirements

Architecture

Assignment



Introduction

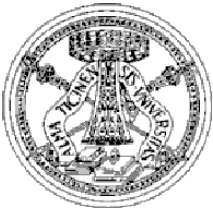
Project Scope

Requirements

Architecture

Assignment

- The project shall impact patients and GPs, while Labs will be almost unaffected.
- In more detail:
 - GPs are critical, since they are unionized and may adverse digital systems
 - Patients are critical, because they have a variable attitude toward apps, depending on their age, culture, language, etc.
- The project, with 10 million of potential users, needs a **pilot district**, which should be small enough but complete.



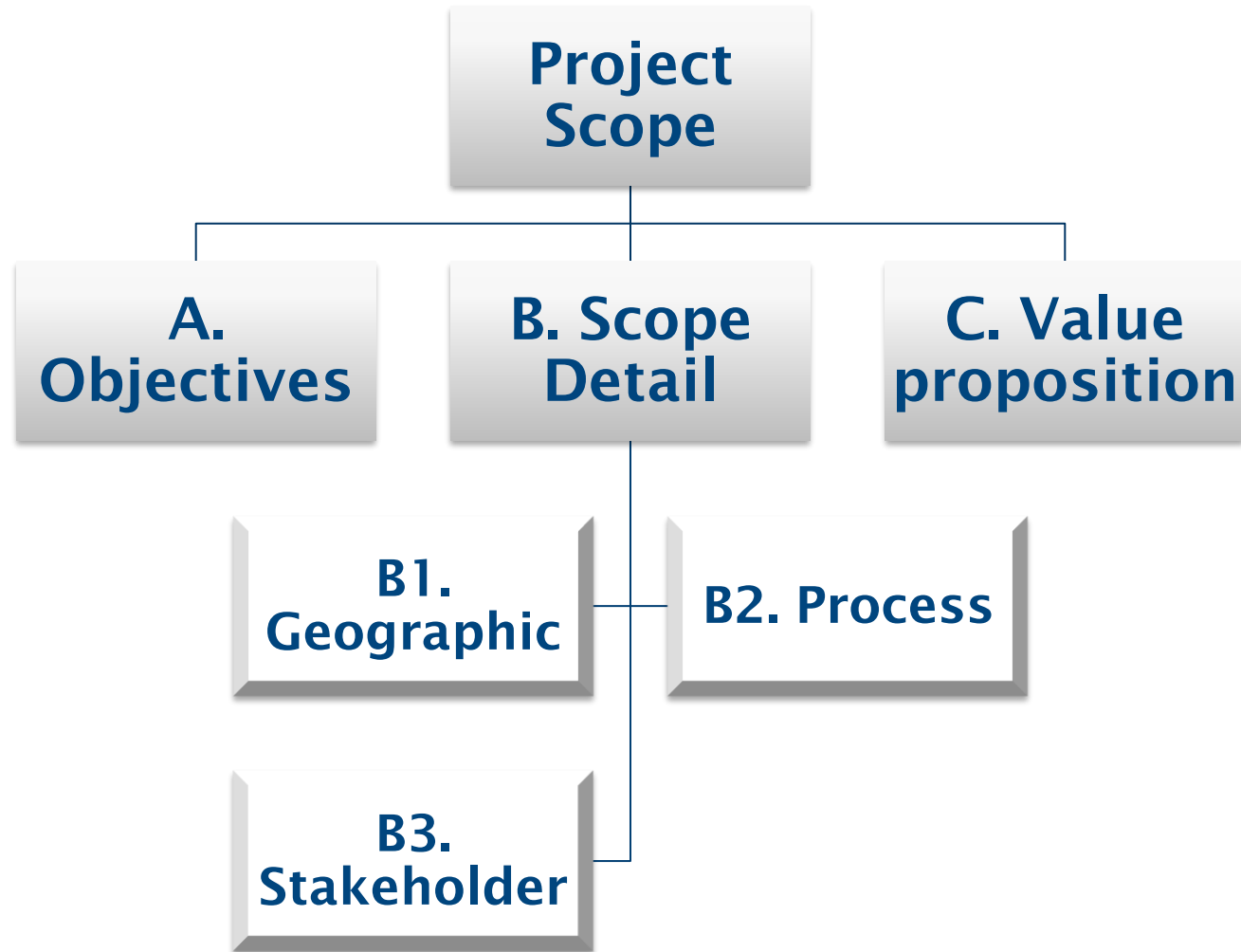
Introduction

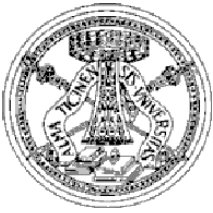
Project Scope

Requirements

Architecture

Assignment





A. Project objective

Introduction

Project Scope

Requirements

Architecture

Assignment

- Project objective is a first section of Project Scope
- It states the results the project should obtain

Minister (1)

Regions (20)

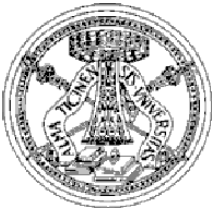
Provinces (200)

Hospitals (+1,000)

Labs (+1.000)

GPs (ca 30K)

Patients (ca 60.000K)



Introduction

- Project scope states what is within the project

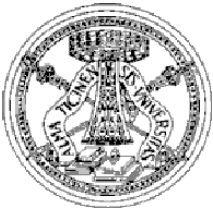
Project Scope

- You should define the scope at multiple levels
 - Geography, i.e., the geographical area the project targets
 - Business Process, i.e., the activities to be supported
 - Stakeholders, i.e., organizations or people who are involved with the business process or system and are critical for the success of the project and system

Requirements

Architecture

Assignment



B.1 Geographic scope

Introduction

Project Scope

Requirements

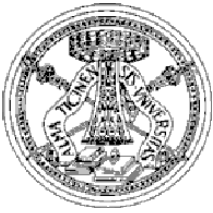
Architecture

Assignment



Region	Capital	Area	Population
Valle Aosta	Aosta	3,263	128,298
Piemonte	Torino	25,402	4,424,467
Liguria	Genoa	5,422	1,583,263
Lombardia	Milano	23,844	10,002,615
Veneto	Venice	18,399	4,927,596
Trentino	Trento	13,607	1,055,934
Friuli-VG	Trieste	7,858	1,227,122
Emilia-Romagna	Bologna	22,446	4,450,508
Toscana	Firenze	22,993	3,752,654
Marche	Ancona	9,366	1,550,796
Umbria	Perugia	8,456	894,762
Lazio	Roma	17,236	5,892,425
Campania	Napoli	13,590	5,861,529
Abruzzo	L'Aquila	10,763	1,331,574
Molise	Campobasso	4,438	313,348
Puglia	Bari	19,358	4,090,105
Basilicata	Potenza	9,995	576,619
Calabria	Catanzaro	15,080	1,976,631
Sardegna	Cagliari	24,090	1,663,286
Sicilia	Palermo	25,711	5,092,080

Geographic scope defines the geographic area the project shall address, i.e., Lombardy



B.2 Business process scope

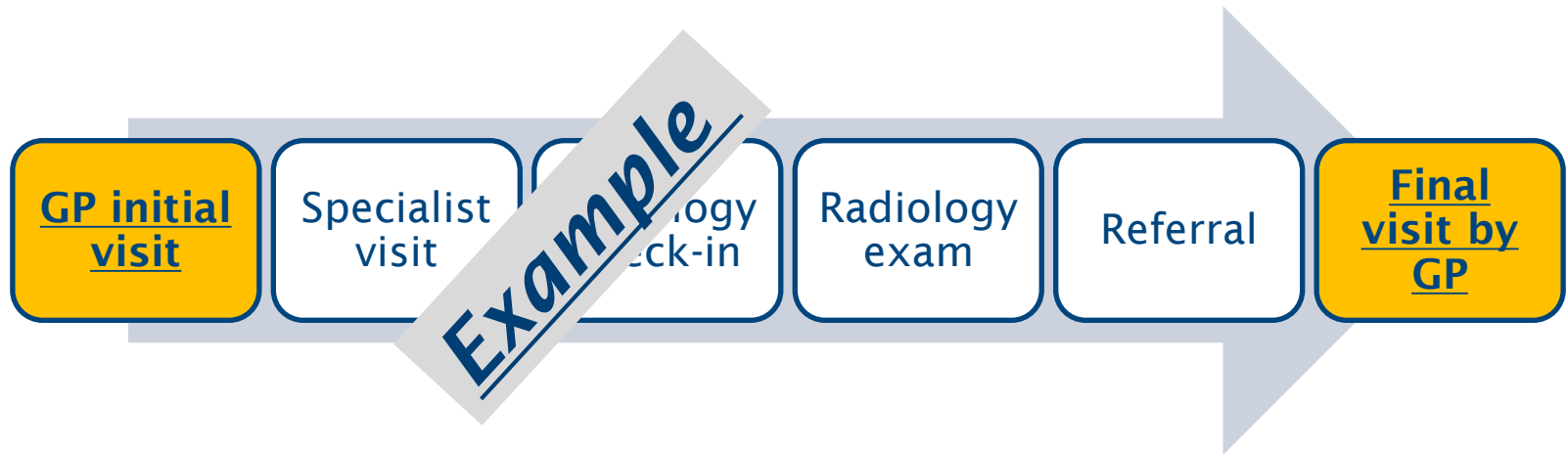
Introduction

Project Scope

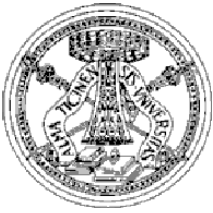
Requirements

Architecture

Assignment



- Business Process scope defines the process tasks the project shall address
- Hence, you should show the whole Business Process and underline the phases your project will address
- In the above example, the project will address the phases filled in orange, e.g., GP initial visit and GP final visit
- Alternatively, the business process can be represented by a CRASO diagram or any context diagram



B.3 Stakeholder scope

What stakeholders are

Introduction



Project Scope



Requirements

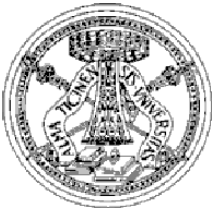


Architecture



Assignment

- Stakeholders are people/organizations who hold an interest (STAKE) in the project
- Without stakeholders, the project cannot be done and/or the system cannot work; therefore, they should be involved and identified early, at the scoping stage.
- Stakeholders can be inside or outside the process, e.g., in HC case study:
 - Region staff are outside stakeholders, who shall be informed on lab exams, because they pay it; also, Region staff define related standards
 - GPs are inside stakeholders, because they perform the visit
- In your assignment you shall identify the stakeholders of your project, based on the categorization shown by the subsequent page.



B.3 Stakeholder scope

Stakeholder categorization

Introduction



Project Scope



Requirements



Architecture



Assignment

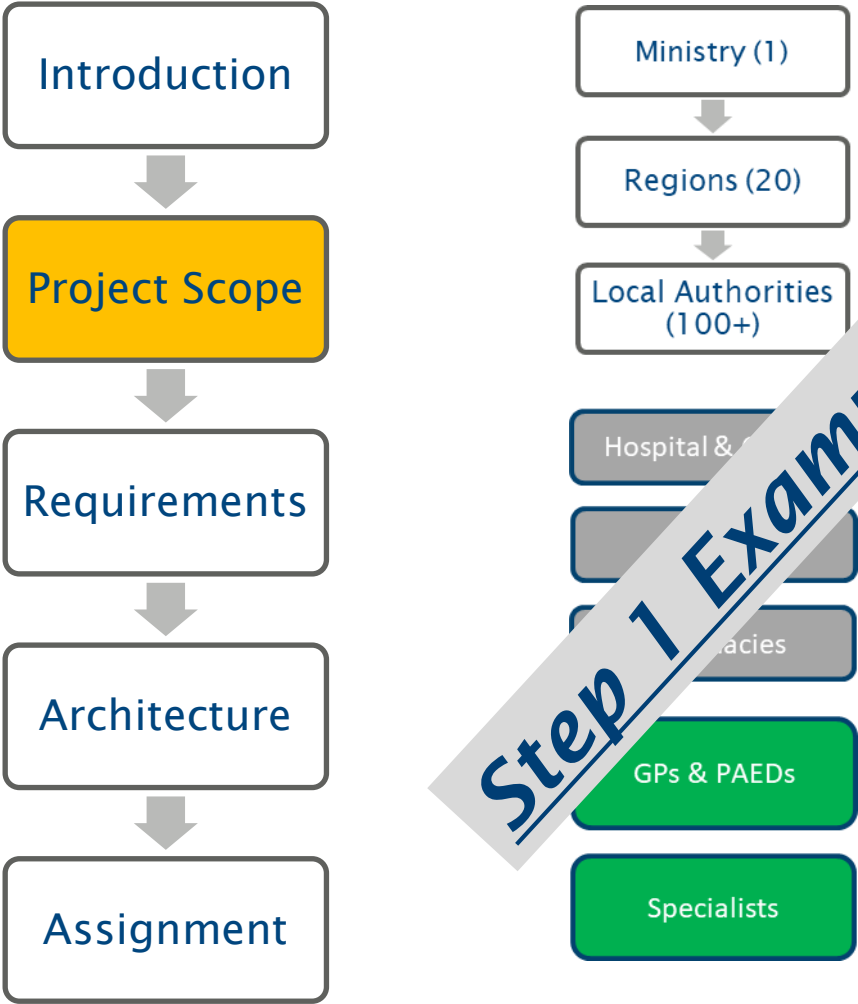
- Stakeholder can be categorized as follows
 - Rulers i.e. People/Organizations who set the rules the process should comply with; e.g.
 - Government and regulatory bodies, which set overall rules e.g., on data privacy;
 - Organization's Information Technology Department (it sets digital standards the system should comply with);
 - Functional managers in the user organizations, who set functional rules, e.g., reporting policies;
 - Operators i.e. People/Organizations who work in the process but do not interact with the system e.g. hospitals in a patient information system;
 - Users, i.e., people and organizations who shall use the system, e.g., passengers in an airline reservation system.



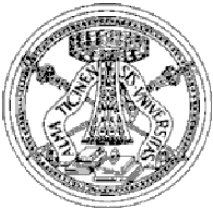
B.3 Stakeholder scope

Stakeholder identification

- To identify stakeholders, we suggest these steps:
 - Analyze the organizations/roles involved in the project;
 - Fill a stakeholder grid where you list the stakeholders who play a role (i.e. Ruler, Operator, User) and comment your involvement.



Stakeholder	Role	Comment
Ministry	Ruler	x
Regions	Ruler	xxxxxxx
Local Authorities	Ruler	xxxxxxxxxx
Hospitals & Clinics	Operator	xxxxxxxxxxxxxx
Labs	User	Check-in, Exam, etc.
Pharmacies		Access
GP		Prescriptions etc.
Specialist	er	xxxxxxxxxxx
Patient	User	Prescriptions etc.



Introduction



Project Scope



Requirements

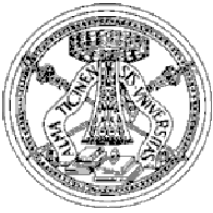


Architecture



Assignment

- Value propositions describe the value each stakeholder class can obtain from the system, i.e., the value added by the system;
- Value propositions can be written in plain English;
- We here exemplify value propositions for the **GP visit.**



C. Value propositions: GP visit example

Value proposition example

Introduction



Project Scope



Requirements



Architecture



Assignment



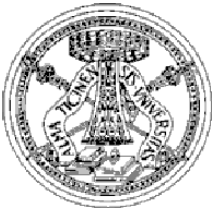
- Can easily check the availability at his /her GP
- Can manage his/her visit reservation on any channel
 - Mobile
 - Web
 - Call Center
 - Authorized Counter
- *Can access his/her medical history (EPR)*



- Can access reservations
- *Can access the patient's medical history (EPR)*
- Can add confidential information on the patient
- Can check availability/book on the agenda of specialist, labs, hospitals
- Prescriptions are automatically added to the Regional EPR



- Manages reception procedures
 - identification of patients
 - ticket payment (if applicable)
 - emergency visit
 - etc.
- Can access at any moment reservations
- Can check availability/book on the agenda of specialist, labs, hospitals
- **Cannot access EPR**



Introduction



Project Scope



Requirements

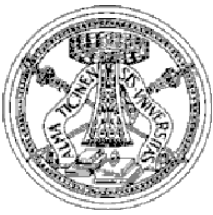


Architecture



Assignment

- Requirements state what the system should do:
 - Functional Requirements
 - What a system should do regardless its implementation
 - Non-Functional Requirements
 - Response time and alike performances
 - Security Requirements address data privacy (see GDPR), data protection, backup, downgrading, etc.
- Functional requirements can be defined by
 - Agile paradigm (Theme/Epic/User Story)
 - Abstract Modeling Languages (CRASO/Hierarchical Diagram /Flow Diagram; UML)
 - Hybrid: Abstract Modeling + Agile



Introduction

Project Scope

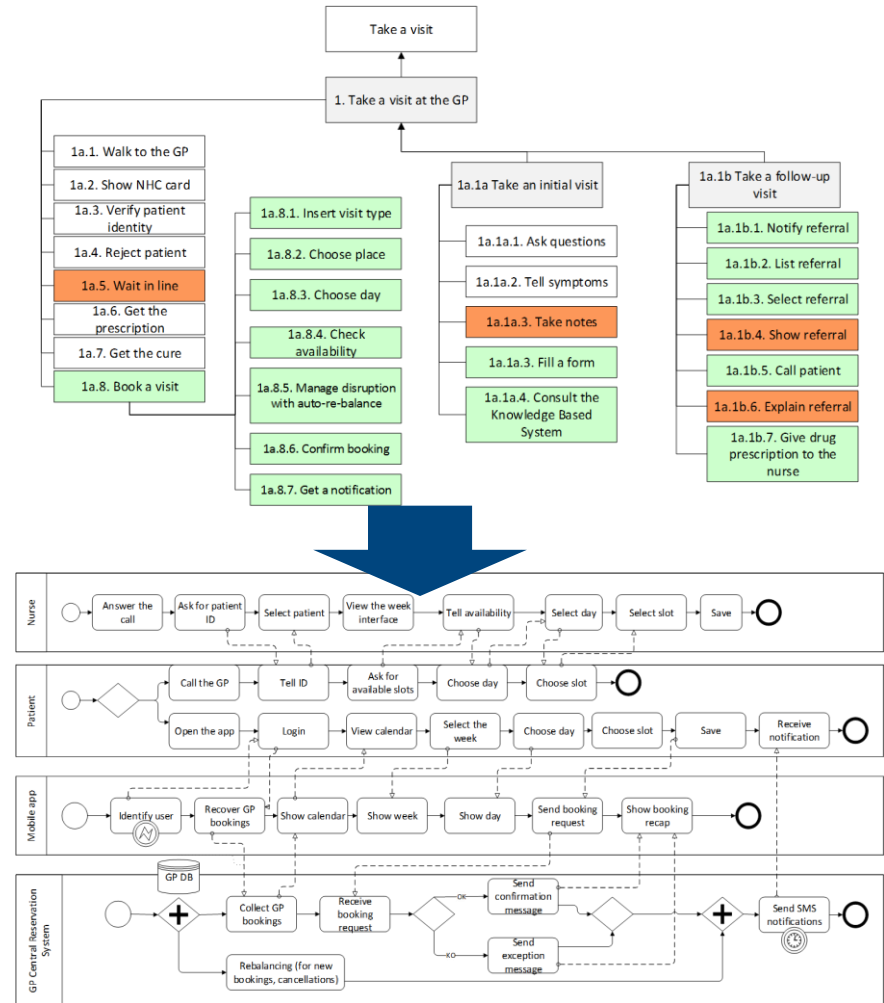
Requirements

Architecture

Assignment

- Functional requirements should compare
 - As-is = current situation
 - To-be = what the system shall accomplish
- So, functions can be:
 - Unchanged (white)
 - Changed (orange)
 - New (green)
 - Discontinued (orange)
- In our approach, Functional Requirements are modeled in 2 steps
 - Hierarchical diagram, define the structure backlog
 - Flow diagram to show dynamics of the system and dependencies between functions

A. Functional Requirements: Abstract Modeling Option





A. Functional Requirements: Abstract Modeling Option: H-diagram

Introduction

Project Scope

Requirements

Architecture

Assignment

Analysis goes down to elementary activities / tasks

Take a visit

1. Take a visit at the GP

- **White** elements are unchanged
- **Orange** elements are to be deleted or changed
- **New** elements are **green**

1a.1. Walk to the GP

1a.2. Show NHC card

1a.3. Verify patient identity

1a.4. Reject patient

1a.5. Wait in line

1a.6. Get the prescription

1a.7. Get the cure

1a.8. Book a visit

1a.8.1. Insert visit type

1a.8.2. Choose place

1a.8.3. Choose day

1a.8.4. Check availability

1a.8.5. Manage disruption with auto-re-balance

1a.8.6. Confirm booking

1a.8.7. Get a notification

1a.1a Take an initial visit

1a.1a.1. Ask questions

1a.1a.2. Tell symptoms

1a.1a.3. Take notes

1a.1a.3. Fill a form

1a.1a.4. Consult the Knowledge Based System

1a.1b Take a follow-up visit

1a.1b.1. Notify referral

1a.1b.2. List referral

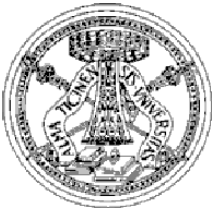
1a.1b.3. Select referral

1a.1b.4. Show referral

1a.1b.5. Call patient

1a.1b.6. Explain referral

1a.1b.7. Give drug prescription to the nurse



A. Functional Requirements: Abstract Modeling Option

Introduction

- *The NEW process avoids that patients waste time in line at GP office.*
- *The patient can book by phone or app.*
- *In both cases the GP booking system stores and updates the appointment timetable and sends a SMS reminder to the patient*
- *A message informs the patient if the reservation must be canceled because of a conflict.*
- *The GP can immediately understand the patient history by accessing the EPR*
- *The GP enters the actual symptoms and diseases in a standardized form and updates EPR*
- *The GP gets suggestions on cures and exams from the Knowledge Base System*
- *The GP can book a specialist visit on behalf of the patient. If the patient decides to book by himself, the GP can see the status of the prescription and the booking anyway, because the bookings are inserted in a centralized system*
- *Innovations in Follow-up visits include :*
 - *The GP receives the referral from the referral system*
 - *GP and patient communicate by phone, if needed.*
 - *Drug prescriptions are created and given to the nurse that delivers them to the patient*

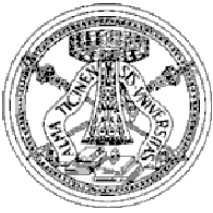
Project Scope

Requirements

Architecture

Assignment

The requirement models should be supported by comments that explain the objective and rationale of the flow design



Execution architecture to-be

Introduction

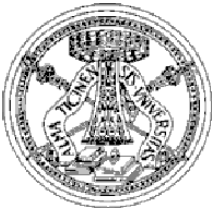
- Systems architecture shows relations between the building blocks of the system.
- In the overall HC project, the to-be architecture shares data and procedures across the systems that are involved in providing HC services; indeed, the HC system is a network of cooperating systems.
- Such integration/collaboration, given the diversity of participating systems can/must rely on:
 - Extranet, i.e., *a controlled private network that allows access to partners. It is like a DMZ in that it provides access to needed services for authorized parties, without granting access to an organization's entire network.*
 - *Virtual shared database – i.e., the access of data spread across the network though hyperlinks – needed when data are images and privacy requirements are compelling.*

Project Scope

Requirements

Architecture

Assignment



Execution architecture to-be

Introduction



Project Scope



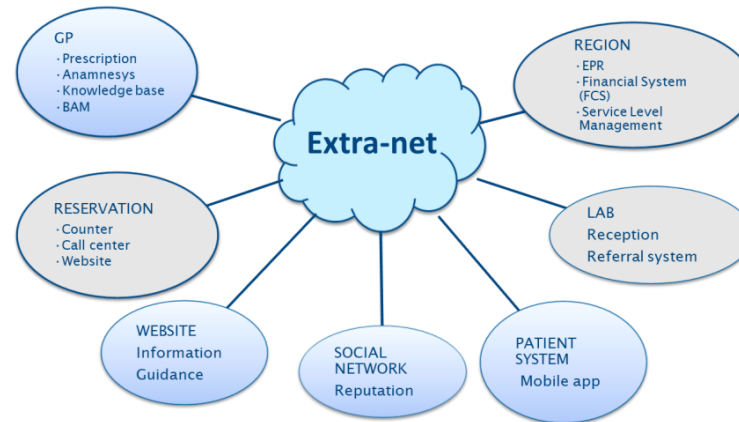
Requirements



Architecture



Assignment



- The overall project includes architecture developments at various levels
- Apps: GP, Website, Reputation, Patient
- Infrastructure: Extranet
- Extranet enable systems to cooperate
- Specifically, students should detail the architecture of their specific assignment



Execution architecture to-be

Introduction



Project Scope



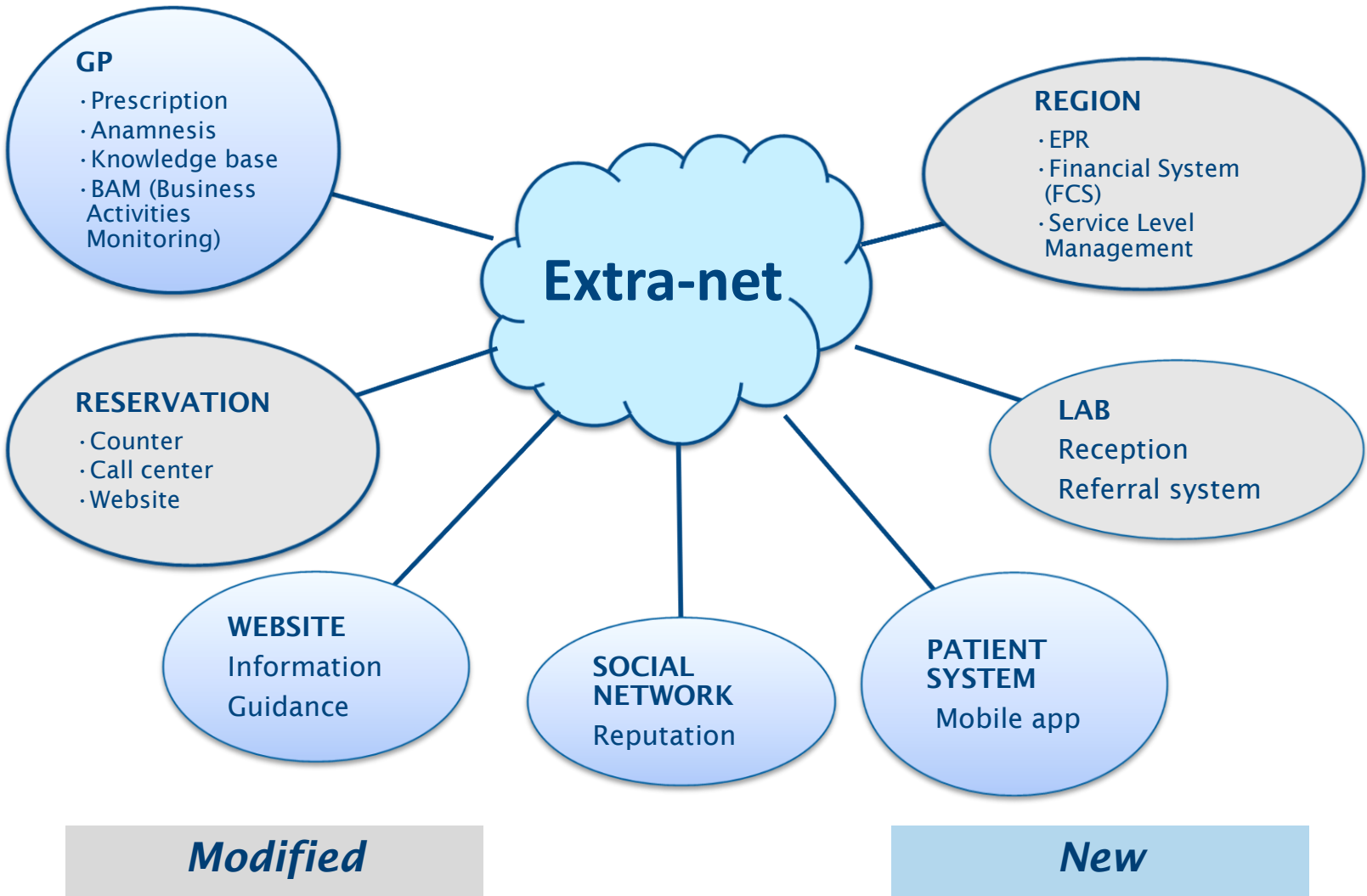
Requirements



Architecture



Assignment





Execution architecture to-be

Introduction

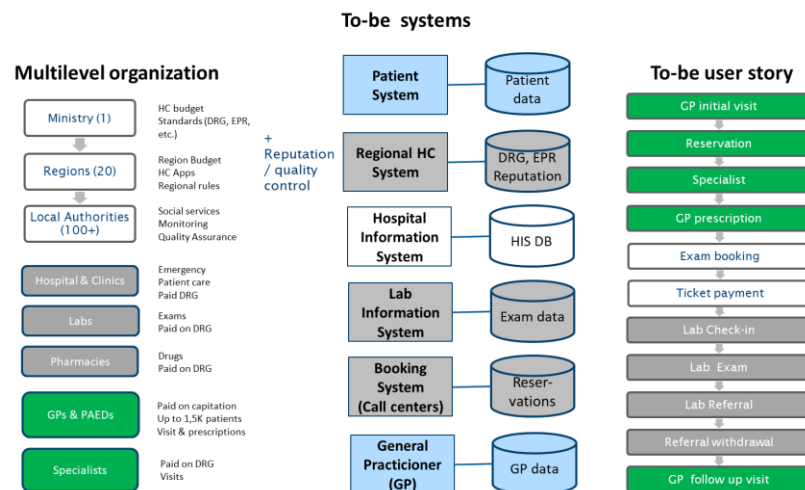
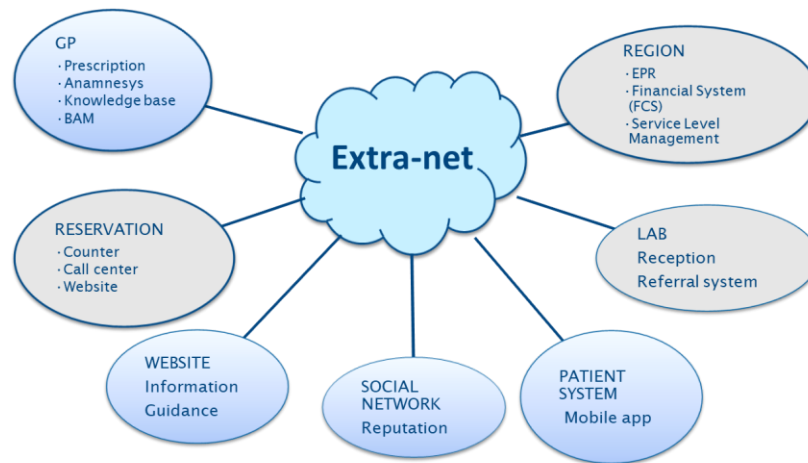
Project Scope

Requirements

Architecture

Assignment

- Process / things added :
 - Patient's mobile / wearable system
 - Standard lab booking platform (= HC operators agenda management)
 - Standard GP booking (= GP agenda management)
 - AI for the doctor
 - DW and analysis models for the Region
 - Etc.



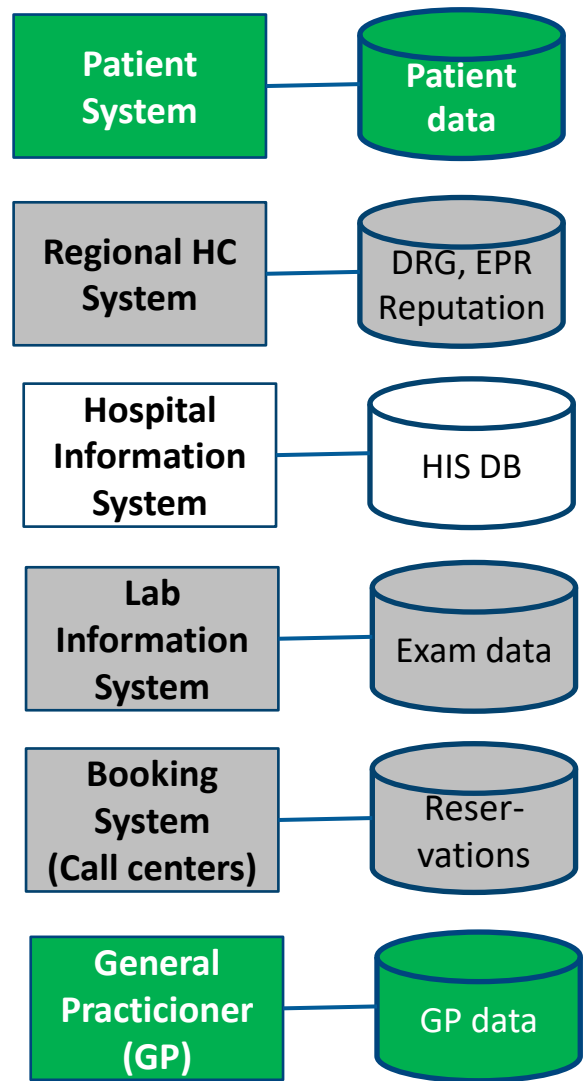


Multilevel organization



+ Reputation / quality control

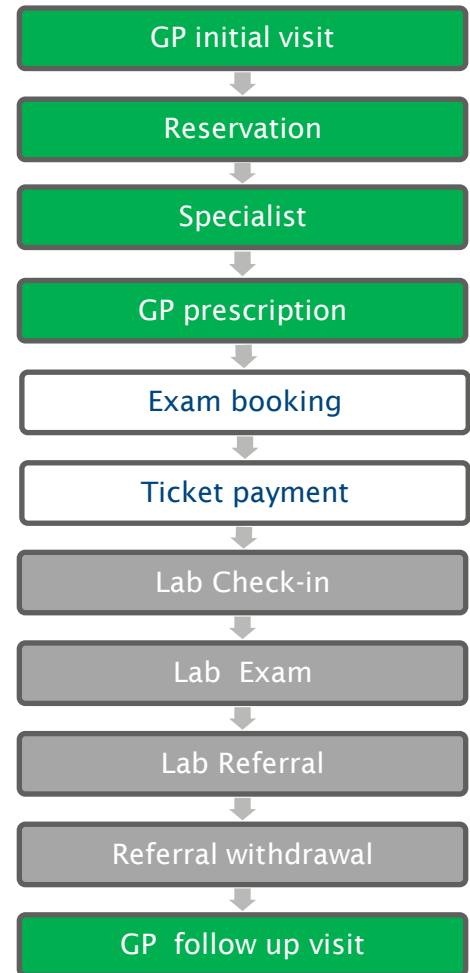
To-be systems



NEW

MODIFIED

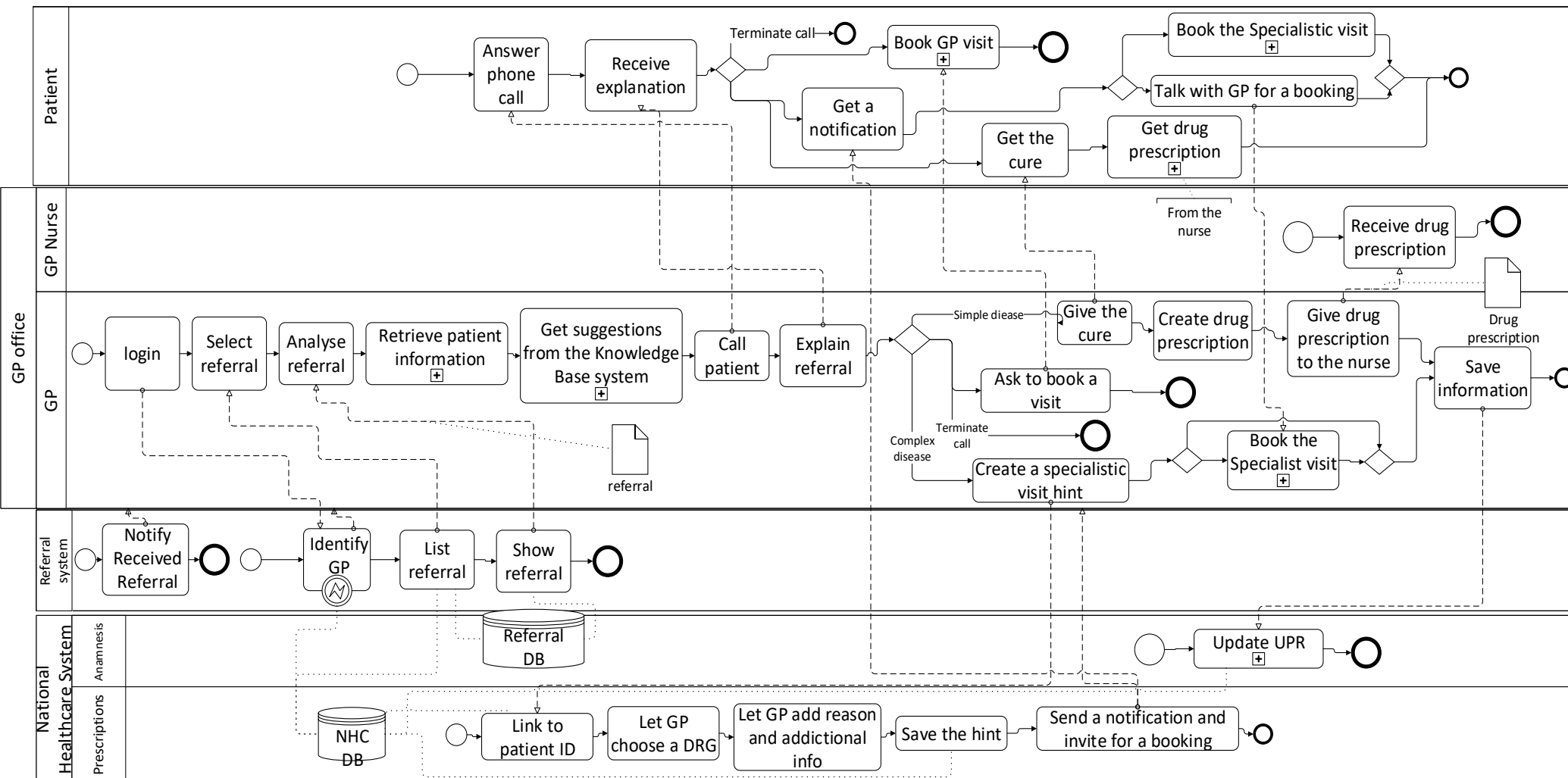
To-be process





A. Functional Requirements: Abstract Modeling

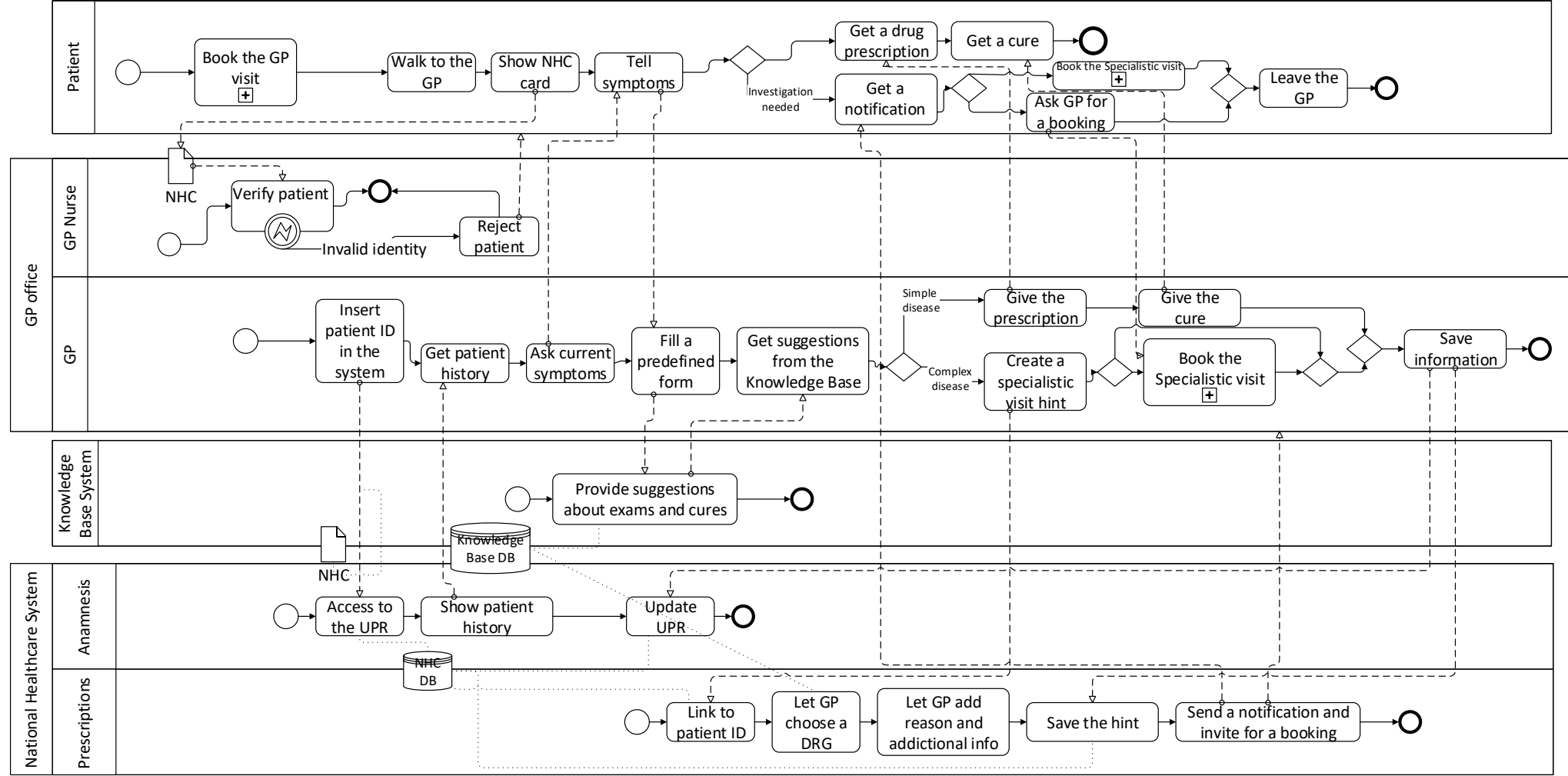
Flow diagram : follow-up visit





A. Functional Requirements: Abstract Modeling

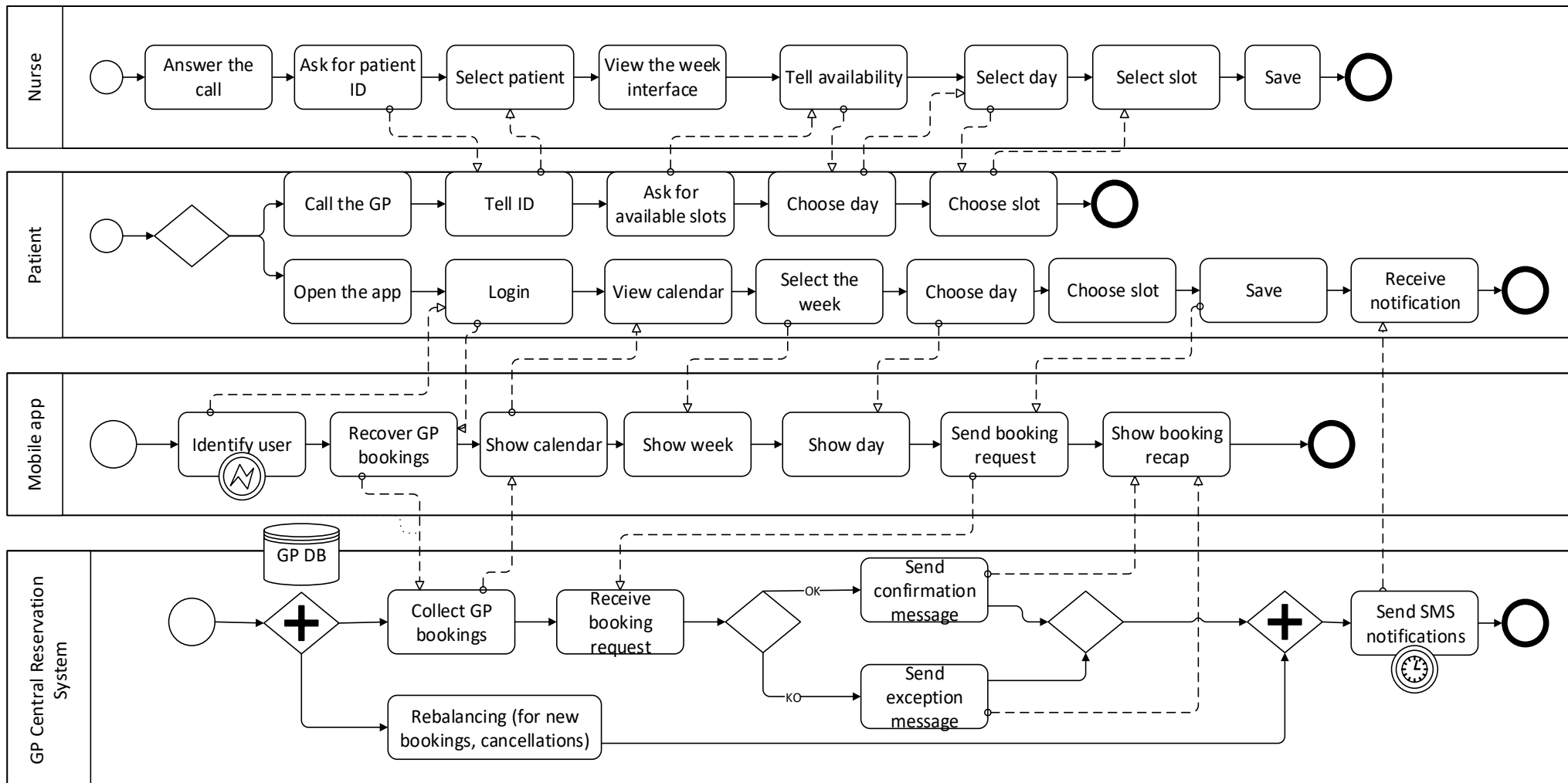
Flow diagram: Book a visit

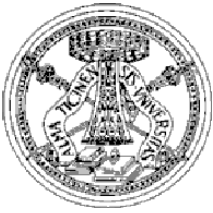




A. Functional Requirements: Abstract Modeling

Flow diagram : Initial visit





Introduction

Project Scope

Requirements

Architecture

Assignment

• Patient's App

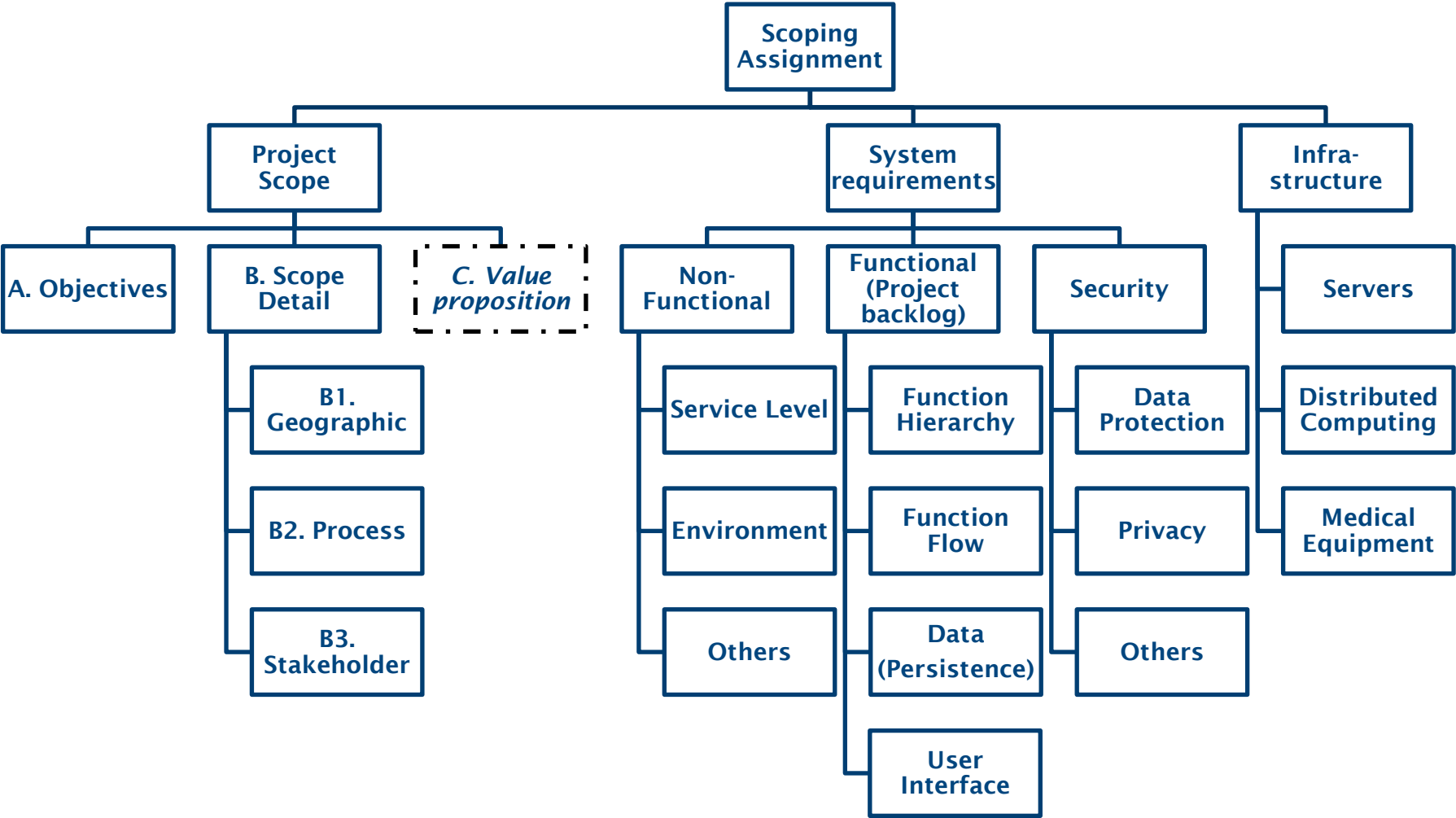
- Runs on Apple, Android, Microsoft Desktop, Apple Desktop
- Is multilingual: Italian, English, Chinese, Arabic
- Supports the full cycle of healthcare from the patient viewpoint
 - Family Doctor: selection/change, visit reservation
 - EPR/HER: access to all documents (prescriptions, discharges, lab reservations, referrals, waiting list, emergency calls, etc.)
 - Reservations for specialists, lab and operations
 - Emergency calls

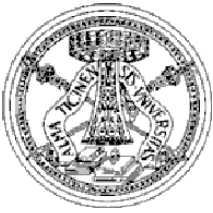
• Lab Management

- Planning and scheduling of equipment and medical staff
- Patient identification with / without ID
- Ticket management
- Acceptance /check-in of patients
- Workload management (assignment of patients to doctors and exam rooms)
- Equipment and exam rooms management
- Management of exam lines (exam number)
- Sample management
- Referral support
- Radiography management



Assigment structure





Introduction



Project Scope



Requirements

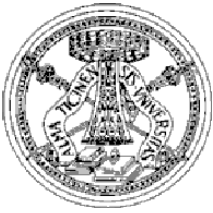


Architecture



Assignment

- The assignment:
 - Addresses a specific scope
 - Is a PPT presentation, with
 - Cover page:
 - Name and ID of team members;
 - Assignment release
 - Pages: Author (each team member can author a different page)



Introduction



Project Scope



Requirements



Architecture



Assignment

- Scope: the assigned project scope should be covered
- Completeness: all the assignment sections should be developed
- Correctness: diagrams and assignment structure should be well formed and not violate relevant syntax rules
- Readability
 - Correct English
 - Visible colors (no Yellow, no Light-Green)
 - Accessible font size